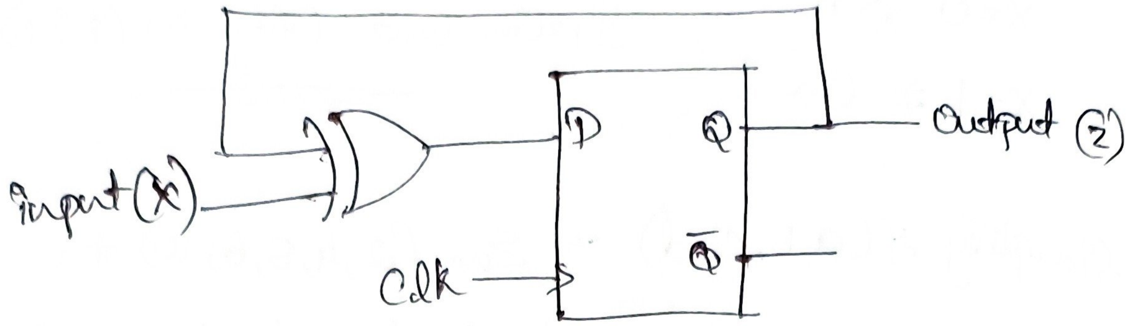
Day 4 Assignment - Sequential

1) Explain the behaviour of the given circuit.



XOR gate inputs are

External input = X

Feedback input = Q

$D = X \oplus Q$

$Q_n = D$

$Q_n = X \oplus Q$

$Z_n = X \oplus Z.$

State table

Present State (Q)	Input (X)	Q_n.
0	0	0
0	1	1
1	0	1
1	1	0

Case 1 : $X = 0 \Rightarrow Q_n = Q$

Case 2 : $X = 1 \Rightarrow Q_n = \overline{Q}$

The circuit behaves like a T-flip flop.

2) Design a posedge triggered D-FF using 2:1 MUX only. Master latch ⇒ $M = \overline{Clk}\,D + Clk\,M$.

Slave latch ⇒ $Q = Clk\,M + \overline{Clk}\,Q$

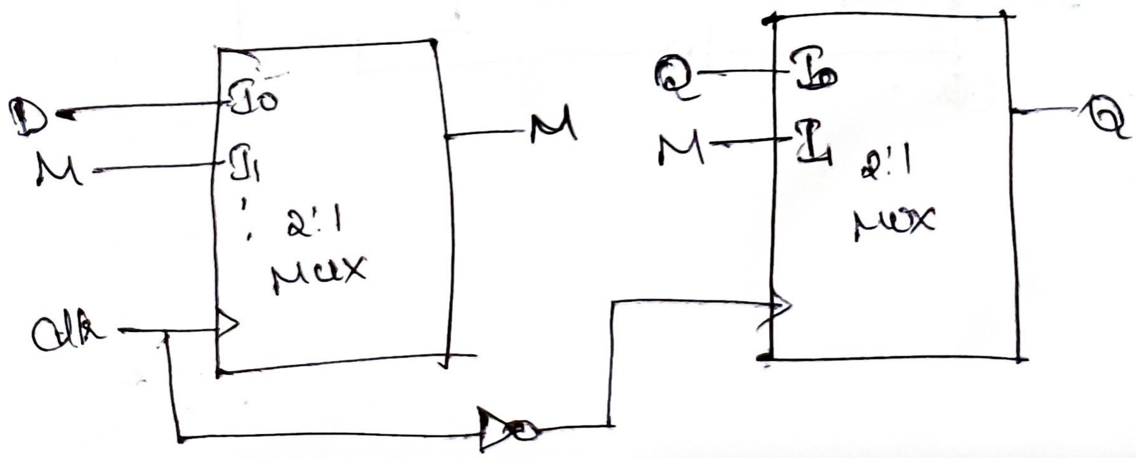
~~Clk = 0~~.

Clk = 0.

MUX 1 input	Connection.
I_0	D
I_1	M (feedback)
Select	Clk
Output	M.

Clk = 1

MUX 2 input	Connection.
I_0	Q (feedback)
I_1	M
Select	clk
Output	Q.



3) Design a negedge Triggered T-FF using 2:1 MUX only characteristics equation of T-flipflop is $Q_n = T \oplus Q$

For 2:1 MUX

Select line - T-flipflop

$I_0 = Q$

$I_1 = \overline{Q}$

$m = \overline{clk}\,m + clk(T \oplus m)$

$Q = clk\,Q + \overline{clk}\,(m)$

$Y = \overline{T}Q + T\overline{Q}$

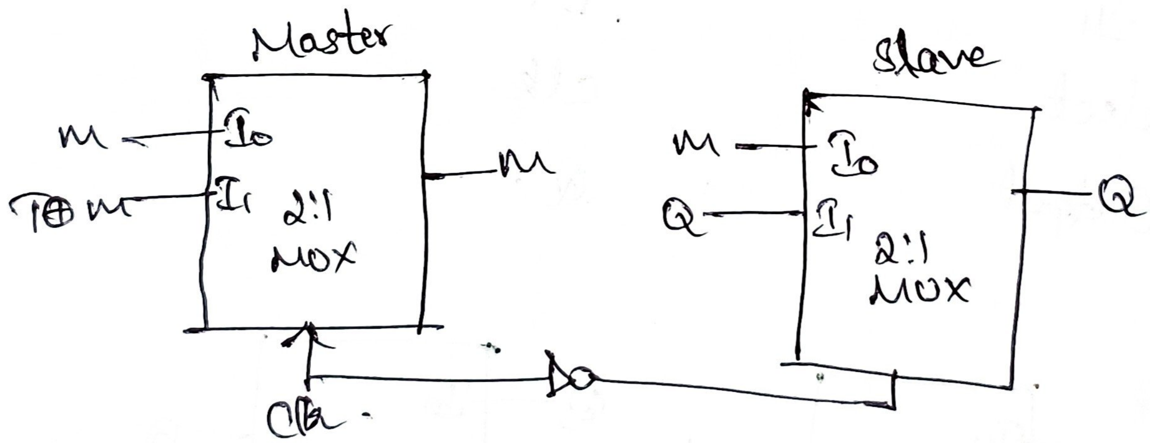
$Y = T \oplus Q$

Master latch $clk = 1$

Input	Connection.
I_0	m (feedback)
I_1	$T \oplus m$
Select	clk
Output	m.

Slave latch, $clk = 0$

Input	Connection.
I_0	m
I_1	Q (feedback)
Select	clk.
output	Q



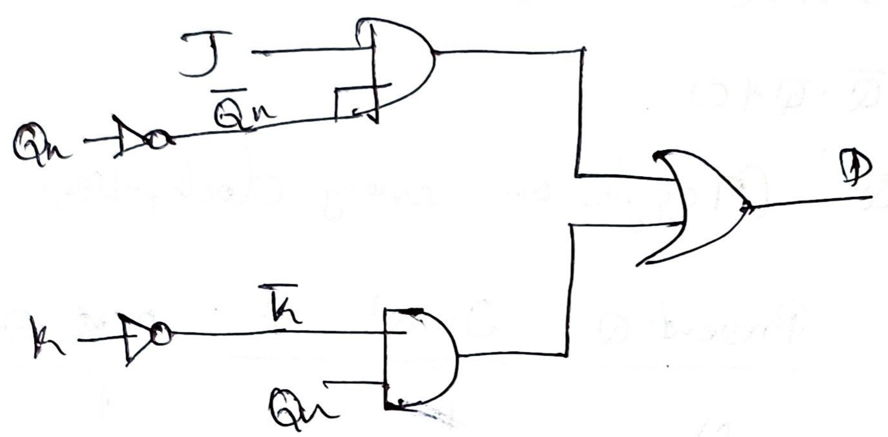
4) Convert by adding external gates:

a) A D-flipflop to a JK flip-flop.

$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$

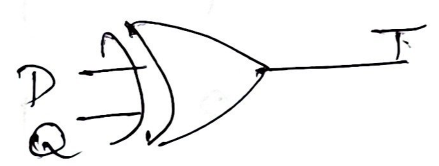
$Q_{n+1} = D$.

$D = J\bar{Q}_n + \bar{K}Q_n$.

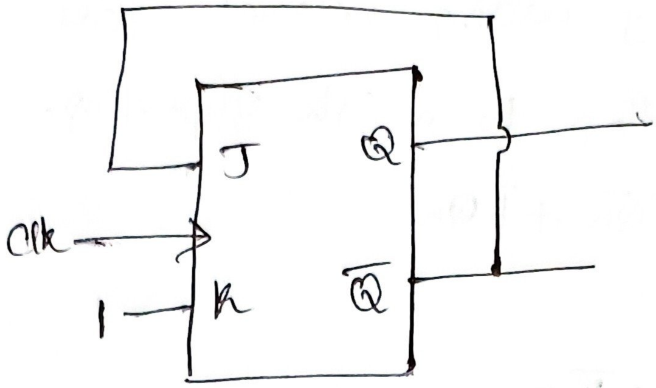


b) A T-flipflop to a D-flipflop, $Q_{n+1} = T \oplus Q$

$Q_{n+1} = D \Rightarrow D = T \oplus Q$, $T = D \oplus Q$.



5) In a positive edge triggered JK flipflop, we have $J = Q'$ and $K = 1$ as shown in the figure below. Assume the flip flop was initially cleared and then clocked for 6 clock pulses, find the sequences at the output for these 6 clock cycles.



$$Q_{n+1} = J\overline{Q_n} + \overline{K}Q_n.$$

$$K = 1 \Rightarrow \overline{K} = 0 \qquad J = \overline{Q}$$

$$Q_{n+1} = \overline{Q}\cdot\overline{Q} + 0$$

$$Q_{n+1} = \overline{Q}$$ (Toggles on every clock pulse)

Clock pulse	Present Q	J = Q'	K	next Q.
0	0	1	1	1
1	1	0	1	0
2	0	1	1	1
3	1	0	1	0
4	0	1	1	1
5	1	0	1	0
6	0	1	1	1

Sequence is $0 \to 1 \to 0 \to 1 \to 0 \to 1 \to 0$